

it. It is not an independent addressable entity, and is not shared across multiple computers. Typically, this is known as 'Direct Attached Storage' (DAS). Access to the data in secondary storage is through the server to which it is attached; hence, if the server is down due to some failure, the data becomes inaccessible. Also, as data storage requirements increase, we need to have greater storage capacity. We produce 2.5 quintillion bytes of information every day, out of the Web searches we do, the online purchases we make, the mobile calls we make, and the social network presence we have (a quintillion is $1000 \times 1000 \times 1000$ times a billion). Given the volume of Big Data that gets produced, the storage requirements go on increasing exponentially. However, in case of Direct Attached Storage, the number of I/O cards (for example, SCSI cards) that can be connected to a computer is limited. Also, the maximum length of a SCSI cable is 25 M. Given these restrictions, the amount of storage that can be realised using the conventional directly attached storage is limited. Also note that DAS results in uneven storage utilisation. If one of the servers has used up all its disk storage and needs further storage, it cannot use any free storage available in the other servers. This is shown in Figure 1, where disks on Server 2 are full, whereas there is excess disk capacity available on Servers 1 and 3. However, it is not possible for Server 2 to use them.

As opposed to the traditional server-centric paradigm we have seen above, in a storage-centric view, storage exists as an independent entity, apart from the compute servers. Storage can be addressed independently, from multiple servers. A simple form of a storage network is shown in Figure 2, where we can visualise the SCSI cables of the computers having been replaced with connections to the network storage. Though the storage is an independent entity on a network, to the operating system running on the compute server, it appears as if locally attached to the compute server.

Two popular forms of storage-centric architectures are Network Attached Storage (NAS) and Storage Area Network (SAN). The latter (SAN) allows storage entities to exist in a network, which can be accessed from compute servers using either special protocols such as Fibre-Channel or standard TCP/IP protocols such as

iSCSI (internet SCSI). SANs provide block-level access to storage, just like traditional locally attached storage. In contrast, in the case of Network Attached Storage or NAS, a dedicated storage computer exists as an entity on the network, and is accessible from multiple compute

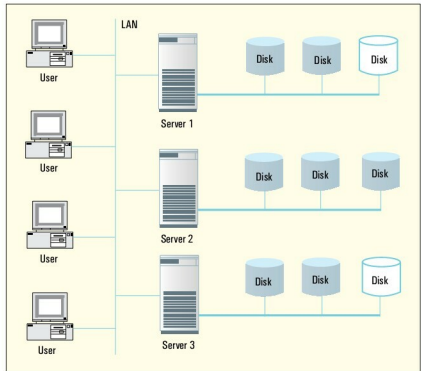


Figure 1: Disks on Server 2 are full

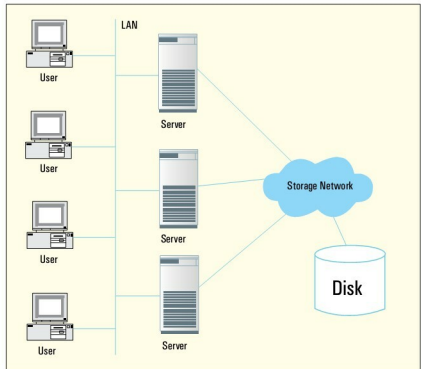


Figure 2: Storage network